

Exploratory Data Analysis Report

The AI-Driven High School Data Intelligence Platform:
Coverage, Structure, and the Rigor Construct in Practice

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Overview

This report covers exploratory data analysis on the platform’s master dataset, identifies the data gaps it exposes, assesses the risks those gaps create, analyzes their impact on the final project, and closes with insights and recommendations. Every number and chart below is computed live from the project’s actual data (`csv_exports/`); nothing here is illustrative or hypothetical. The analysis draws on `benchmarking_v1_2026-07-17.csv` — the final joined artifact, covering 53,966 schools nationwide after the rigor-classification, clustering, and benchmarking passes described in `docs/RIGOR_CLASSIFICATION.md`, `docs/CLUSTERING.md`, and `docs/BENCHMARKING.md` — together with `clustering_v1_2026-07-17.csv` for the clustering-specific section.

A note on this revision: the rigor sections have been explicitly connected to the academic-rigor literature the project’s written report reviews (Sections 2.2, 2.4, and 2.7 of `MSDS_498_version-wk3`). The intent is to show that the five-tier rigor scale is not an ad-hoc construct but a deliberate operationalization of a specific, well-documented body of evidence — and that this EDA puts several of that literature’s central claims to an empirical test on the project’s own data. Works are cited author–year below; full bibliographic details live in the written report’s Section 2 and bibliography.

The environment used to produce these results is standard: `pandas` and `NumPy` for the tabular work, and `Matplotlib` with `Seaborn` for the figures.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set_theme(style="whitegrid", palette="deep")
plt.rcParams["figure.dpi"] = 110

df = pd.read_csv("../csv_exports/benchmarking_v1_2026-07-17.csv", low_memory=False)
cluster_df = pd.read_csv("../csv_exports/clustering_v1_2026-07-17.csv", low_memory=False)
```

```
Main dataset:      34,392 rows x 67 columns
Clustering dataset: 34,392 rows x 72 columns
```

1 Dataset Overview

The master dataset consolidates ten public data sources — NCES, Census, CRDC, College Board, IB, ISBE, CPS, and NAEP — plus the client’s own NU organizational export, joined

primarily on CEEB code. The fully-joined universe (`schools_org_all`) contains 53,966 rows: public schools, private schools, and org-only rows with no school-side match. The descriptive statistics in this report use the version restricted to actual public and private high schools with enrollment at or above 30 — the “cleaning freeze” universe of 34,392 rows described in `docs/TEST_PLAN.md` §1. Within that universe the split is 20,620 public schools to 13,772 private, shown in Figure 1.

```
print(f"Analysis universe: {len(df):,} schools")
print(df["sector"].value_counts().to_string())

fig, ax = plt.subplots(figsize=(6, 4))
sector_counts = df["sector"].value_counts()
ax.bar(sector_counts.index, sector_counts.values, color=["#4C72B0", "#DD8452"])
ax.set_ylabel("Number of schools")
ax.set_title("Schools by Sector (public vs. private)")
```

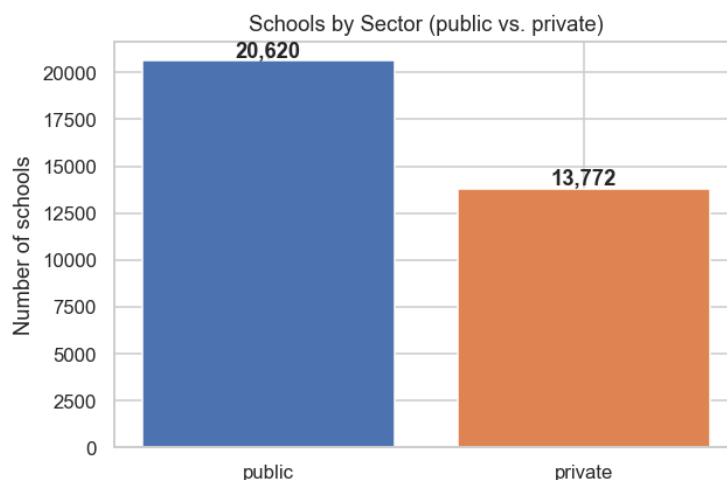


Figure 1: Analysis universe by sector: 20,620 public and 13,772 private high schools (34,392 total).

2 Descriptive Statistics

Table 1 summarizes the core numeric features used across the rigor classification, the funding overlay, and the benchmarking component. Two things are already visible before any modeling: the features carry very different amounts of missingness (from 37% to 68%), and the ones most central to the rigor construct — AP participation and SAT — are among the least complete, a point Sections 3 and 5 return to.

The `frl_rate` maximum of 173.77 is itself a data-quality signal — a free/reduced-lunch “rate” above 100% reflects a known denominator mismatch in the source (enrolled-count

Table 1: Descriptive statistics for the core numeric features (34,392-school universe). “Miss.%” is the share of the universe with a null value.

Feature	Count	Mean	Std	Min	25%	50%	75%	Max	Miss.%
grad_rate_2021	17,420	84.55	17.77	0.50	82.00	91.00	95.00	99.50	49.35
frl_rate	17,230	56.52	26.19	0.12	36.02	54.29	79.07	173.77	49.90
child_poverty_-saip	20,923	15.56	7.00	2.60	10.50	14.80	19.20	60.70	39.16
ap_-participation	12,260	0.19	0.15	0.00	0.08	0.16	0.26	0.78	64.35
testtaker_rate	18,804	0.20	0.15	0.00	0.07	0.20	0.28	0.71	45.32
sat_score_nu	11,098	1148.9	65.17	910.0	1110.0	1150.0	1180.0	1440.0	67.73
per_resident_-child_funding	14,340	16,337	6,291	7,842	11,995	14,845	18,920	41,848	58.30
enrollment_9_-12	21,638	703.4	735.85	30.0	170.0	409.0	1,034	14,517	37.08

versus eligibility-count), documented in docs/DATA_DICTIONARY.md — and is the kind of artifact the coverage and distribution sections below are designed to surface rather than average away.

3 Data Coverage (Missingness) Analysis

Coverage below 100% is not inherently a problem — some fields are legitimately source-specific, such as the Illinois-only ISBE fields — but it is exactly what determines which downstream analyses are trustworthy at what scale, and it is the foundation for the data-gaps discussion in Section 10. Figure 2 ranks the platform’s key features by the share of the universe for which each is populated.

```
coverage = pd.Series({label: df[col].notna().mean() * 100
                        for col, label in coverage_cols.items()}).sort_values()
colors = ["#C44E52" if v < 40 else "#DD8452" if v < 70 else "#55A868"
          for v in coverage.values]
ax.barh(coverage.index, coverage.values, color=colors)
```

The pattern that matters most is not any single bar but where the rigor-relevant signals fall: the AP and test-participation fields that feed the rigor tier sit in the middle band, and the rigor composite itself is populated for under two-thirds of the universe. Section 5 shows that this missing third is not missing at random.

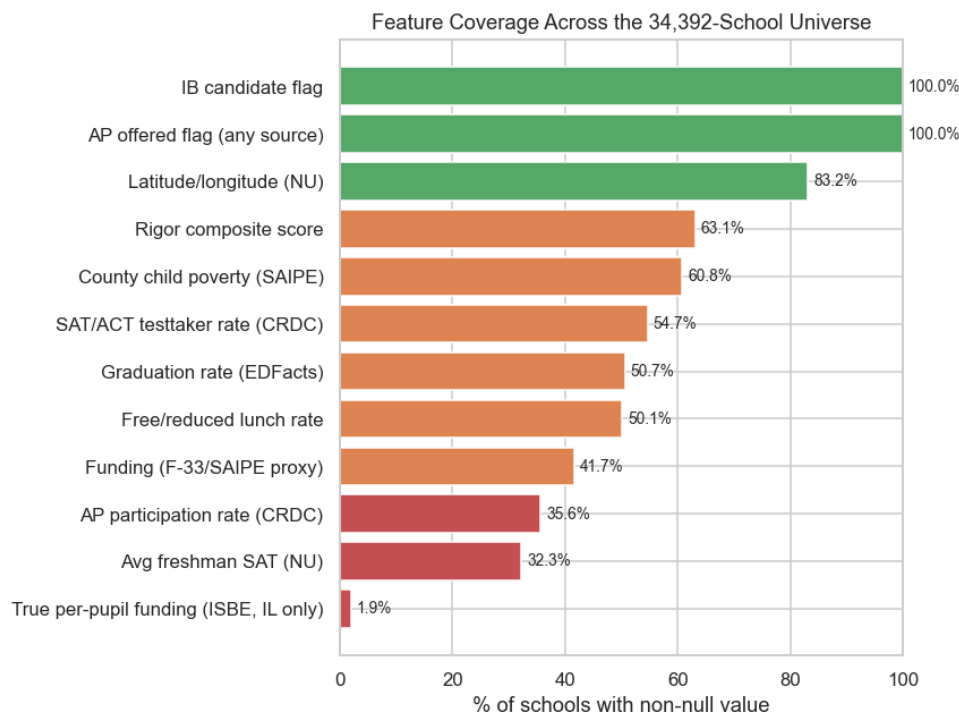


Figure 2: Feature coverage across the 34,392-school universe. Red marks features populated for under 40% of schools, orange 40–70%, green above 70%.

4 Data Distributions

Figure 3 shows the distributions of the key features feeding the rigor classification and the funding overlay. The AP-participation and testtaker-rate distributions are right-skewed and floored at zero, consistent with the offering-count literature discussed in Section 7: most schools cluster at low participation, with a thin tail of high-intensity schools. SAT is approximately normal but narrow (the bulk between 1,100 and 1,200), and the enrollment distribution is long-tailed enough that it is clipped at 3,000 for display.

```
fig, axes = plt.subplots(2, 3, figsize=(15, 8))
sns.histplot(df["sat_score_nu"].dropna(), bins=30, ax=axes[0, 0])
sns.histplot(df["ap_participation"].dropna(), bins=30, ax=axes[0, 1])
sns.histplot(df["per_resident_child_funding_state_local"].dropna(), bins=30, ax=axes[0, 2])
sns.histplot(df["child_poverty_saipe"].dropna(), bins=30, ax=axes[1, 0])
sns.histplot(df["grad_rate_2021"].dropna(), bins=30, ax=axes[1, 1])
sns.histplot(df["enrollment_9_12"].dropna().clip(upper=3000), bins=30, ax=axes[1, 2])
```

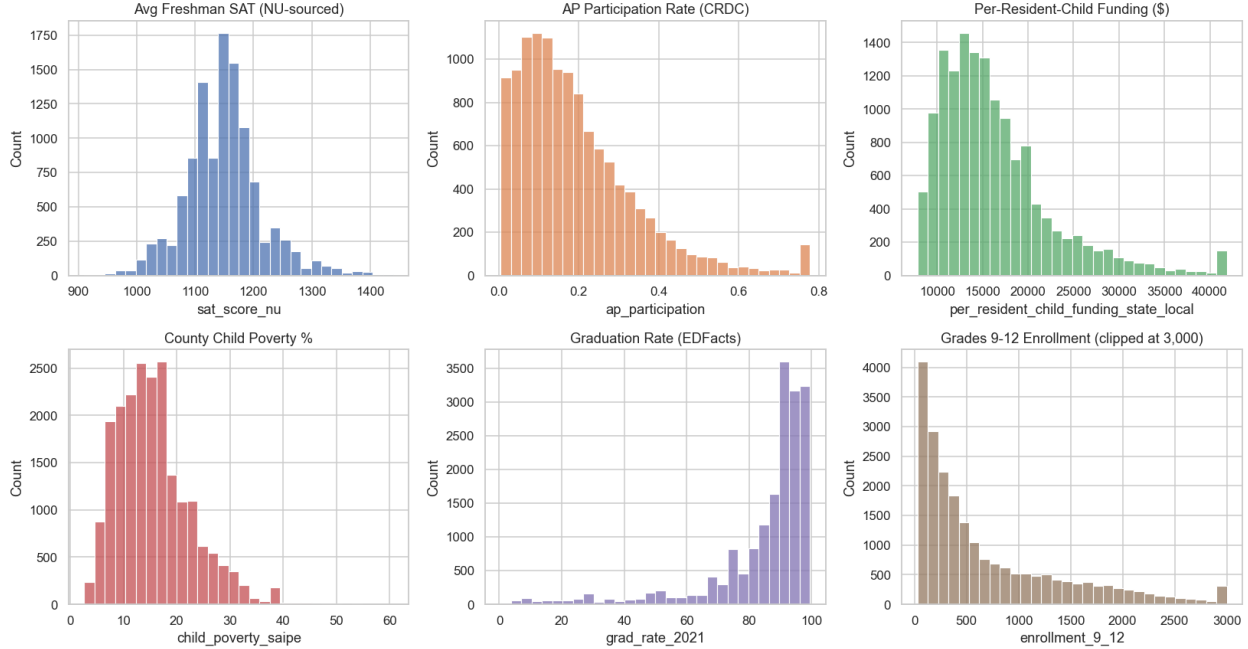


Figure 3: Distributions of the six core features feeding the rigor classification and funding overlay.

5 Public vs. Private: The Real Structural Split

The single most consequential pattern in this dataset is that public and private schools have fundamentally different data *availability*, not merely different values. CRDC — the richest federal source, carrying AP participation, dual enrollment, and testtaker rates — is a public-school-only collection by federal design; private schools depend entirely on what the client’s own NU export happens to carry. Table 2 and Figure 4 make the gap stark: on every CRDC- and EDFacts-sourced feature, private-school coverage is effectively zero, while the only field with full private-school coverage is the NU export flag.

Table 2: Coverage by sector. CRDC- and EDFacts-sourced features are structurally absent for private schools.

Feature	Public %	Private %
AP participation (CRDC)	59.5	0.0
SAT/ACT testtaker rate (CRDC)	91.2	0.0
Grad rate (EDFacts)	84.5	0.0
Funding (F-33 proxy)	69.5	0.0
Has any NU org data	100.0	100.0

This is not fixable by better matching: the data simply does not exist for private schools at the federal level. Its consequence for the rigor construct is direct and is taken up in Section 7 — any rigor tier assigned to a private school rests on a thinner, less-verified foundation than the same tier assigned to a public one.

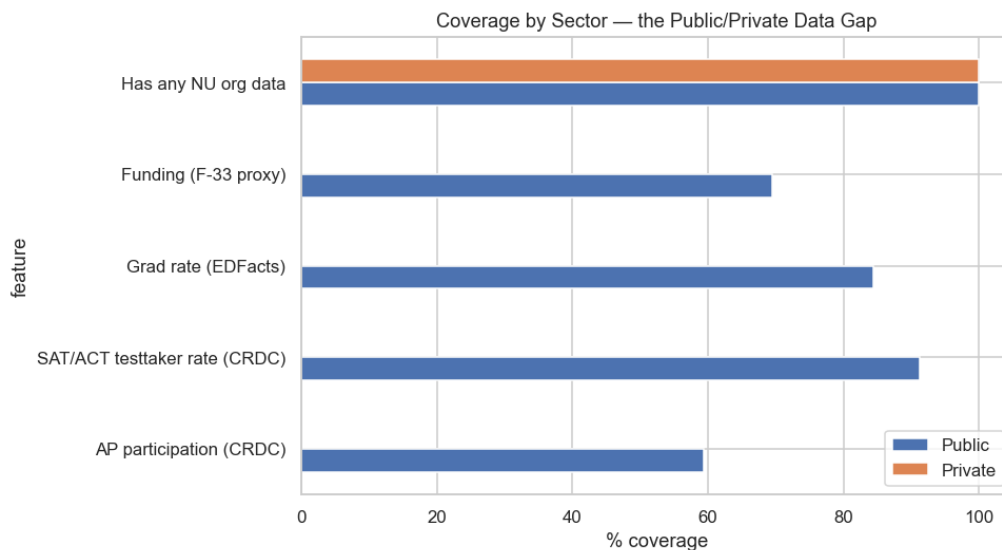


Figure 4: Coverage by sector — the public/private data gap. Every federal source is public-only; private schools rely entirely on the NU export.

6 Correlation Analysis

Figure 5 shows the correlations among the academic, funding, and poverty features that feed the rigor and funding components — the same collinearity that `docs/CLUSTERING.md` §4.3 quantifies via PCA, shown here as a heatmap for direct inspection. The academic-opportunity features move together, and both move with funding and (inversely) with poverty. That structure is exactly what the composite-indicator literature warns about (Section 7): when the inputs to a composite are intercorrelated, the composite’s nominal weights diverge from its effective weights, and an offering-based rigor score risks carrying the socioeconomic signal of its inputs rather than standing apart from it.

```
corr_cols = ["ap_participation", "dual_enrollment_rate", "testtaker_rate",
            "grad_rate_2021", "per_resident_child_funding_state_local",
            "child_poverty_saipe", "latitude", "longitude"]
sns.heatmap(df[corr_cols].corr(), annot=True, fmt=".2f", cmap="RdBu_r",
            center=0, vmin=-1, vmax=1)
```

7 Rigor Classification Summary

The rigor tier is a five-level composite built from AP, CRDC-coursework, and test-participation components; the full methodology, including the nominal-versus- effective weights and the sensitivity analysis, is in `docs/RIGOR_CLASSIFICATION.md`. Figure 6 shows the tier distribution, and the composite is populated for 21,706 of the 34,392 schools (63.1%).



Figure 5: Correlation matrix among academic, funding, and poverty features.

```

tier_order = ["Below Average", "Average", "Demanding",
              "Very Demanding", "Most Demanding"]
tier_counts = df["rigor_tier_label"].value_counts().reindex(tier_order)
ax.bar(tier_counts.index, tier_counts.values, color=sns.color_palette("viridis", 5))
print(f"Scored: {df['rigor_score'].notna().sum():,} / {len(df):,} "
      f"({100*df['rigor_score'].notna().mean():.1f}%)")

```

Scored: 21,706 / 34,392 (63.1%)

Grounding in the rigor literature

The five-tier scale is a deliberate operationalization of the construct the project’s written report adopts (Section 2.2), not an ad-hoc index. Adelman’s two U.S. Department of Education longitudinal studies, *Answers in the Tool Box* (Adelman, 1999) and *The Toolbox Revisited* (Adelman, 2006), established curriculum intensity — not test scores or GPA — as the single strongest predictor of bachelor’s-degree completion, accounting for roughly 43% of variance in the earlier study; that finding is what justifies building a rigor measure from course-offering data at all. The same literature, however, bounds what such a measure can claim. Geiser and Santelices (2004) found that AP and honors course-taking carried little predictive validity for college outcomes once other factors were controlled, while AP *exam performance* did, and the tier here is built almost entirely from availability and participation

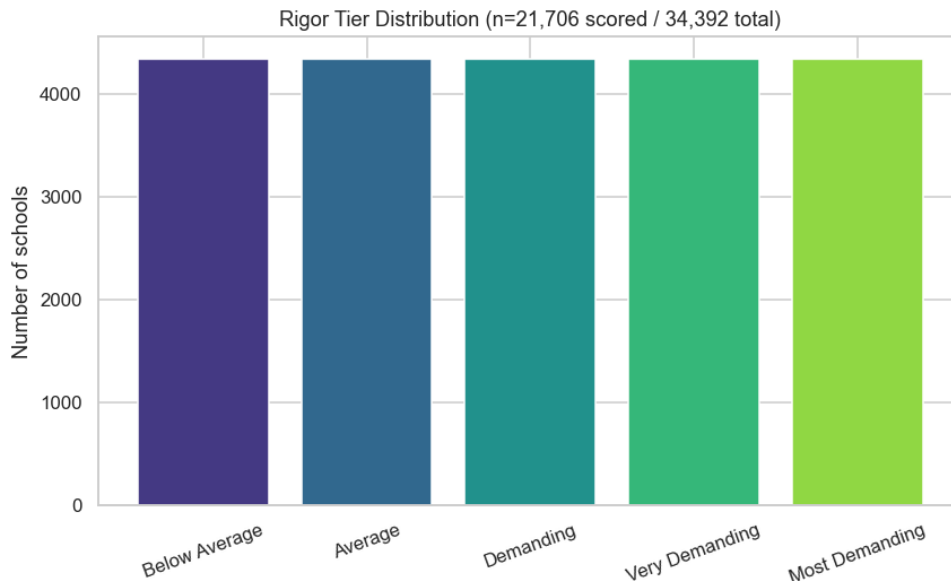


Figure 6: Rigor tier distribution across the 21,706 scored schools.

data — CRDC AP counts, IB flags with no U.S. exam scores, and state AP aggregates — rather than exam outcomes. The tier therefore measures *opportunity structure*, what a school offers, rather than demonstrated academic outcome, and is most honestly read as a “most opportunity-rich” scale rather than a “most demanding” one.

The coverage numbers reinforce that reading. The signal exists for only 63.1% of the universe, and Section 5 showed that the missing 36.9% is concentrated among private schools, not spread at random. Two further strands of the literature explain why the correlation and poverty checks in this report are not optional. Kolluri (2018) documents that low-income students enroll in AP, where it is offered, at under a third the rate of higher-income peers, and that nearly all AP-effectiveness studies are correlational; the U.S. Government Accountability Office (2018) found directly that high-poverty and smaller schools offer fewer college-preparatory courses. A rigor tier built from offering counts will therefore tend to proxy school size and affluence, which means the funding and poverty overlay is not guaranteed to be an independent enrichment layer — it may be collinear with the rigor target itself, exactly the collinearity visible in Section 6’s heatmap. Whether this project’s tier actually inherited that confound is an empirical question, and Section 9 answers it.

8 Clustering Summary

Figure 7 plots the K-means clusters ($k = 4$, selected via the gap statistic) on the first two principal components. The full methodology is in `docs/CLUSTERING.md`, including the honest finding that cluster separation is weak (best silhouette score 0.205, below the conventional 0.25 threshold), shown visually here as overlapping rather than tightly separated point clouds.

Complete- case coverage across all eight clustering features (location, academic, funding, and poverty) exists for only 5,797 schools — 16.9% of the universe — which is itself a finding: the clustering speaks for a minority of well-covered schools, not the whole platform.

```
plot_df = cluster_df.dropna(subset=["pca_component_1", "pca_component_2",
                                     "cluster_kmeans"])
ax.scatter(plot_df["pca_component_1"], plot_df["pca_component_2"],
           c=plot_df["cluster_kmeans"], cmap="Set2", alpha=0.4, s=10)
```

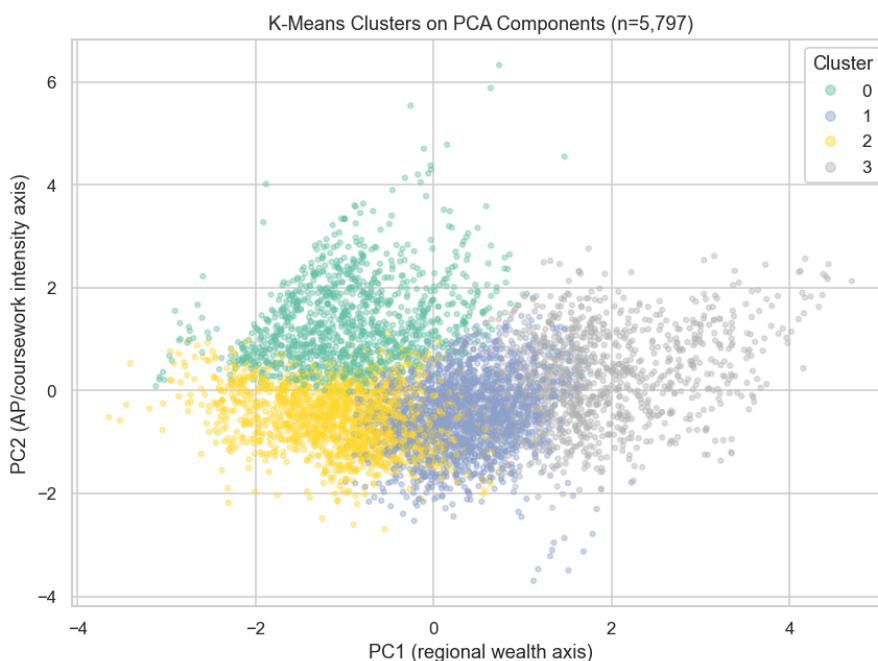


Figure 7: K-means clusters on the first two principal components ($n = 5,797$ complete cases). PC1 reads as a regional-wealth axis, PC2 as AP/coursework intensity.

Complete-case coverage for clustering: 5,797 / 34,392 (16.9%)

9 Performance Benchmarking Summary

Figure 8 shows mean SAT by region and by rigor tier (descriptive only; the full peer-group methodology and the socioeconomic-reproduction finding are in `docs/BENCHMARKING.md`). The decisive numbers are the two rank correlations printed beneath it: SAT achievement correlates -0.385 with county child poverty, while the rigor tier correlates only -0.070 .

```
region_sat = df.groupby("us_region")["sat_score_nu"].mean().sort_values(ascending=
    False)
```

```

tier_sat = df.groupby("rigor_tier_label")["sat_score_nu"].mean().reindex(tier_order)
print(f"spearman(SAT, poverty)          = "
      f"{df['sat_score_nu'].corr(df['child_poverty_saipe'], method='spearman'):.3f}")
print(f"spearman(rigor tier, poverty) = "
      f"{df['rigor_tier_num'].corr(df['child_poverty_saipe'], method='spearman'):.3f}"
      )

```

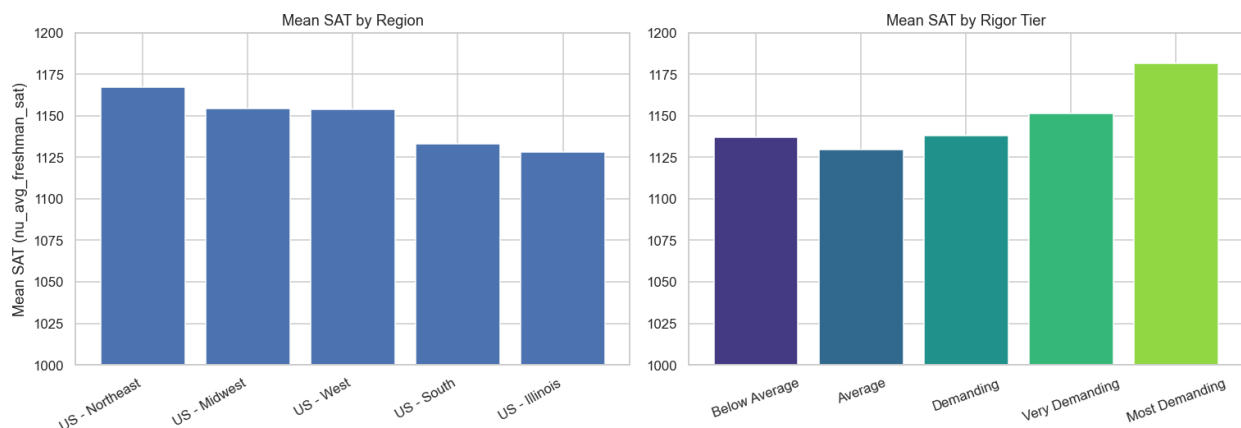


Figure 8: Mean SAT by region (left) and by rigor tier (right).

```

spearman(SAT, poverty)          = -0.385
spearman(rigor tier, poverty) = -0.070

```

This is the composite-indicator literature’s central warning put to an empirical test on the project’s own data. The Stanford Education Data Archive work (Jang and Reardon, 2019; Ho, 2020) established that achievement-*level* indicators are powerfully predicted by socioeconomic composition and therefore reproduce the socioeconomic ordering of American schools — which is precisely what the SAT correlation of -0.385 shows here, an outcome-style measure ranking affluent schools highest and under-resourced schools lowest, the opposite of what the platform’s gap-detection purpose requires. The rigor tier, built from opportunity structure rather than outcomes, does not reproduce that ordering to nearly the same degree (-0.070). The concern raised in Section 7 — that offering counts might smuggle the socioeconomic signal back in — is thus tested and found small in this design, consistent with Bastedo et al. (2022, 2023), who show that contextualized, opportunity-oriented measures improve both equity and predictive accuracy relative to raw scores. The caution that remains is the one Geiser and Studley (2002) and Bettinger et al. (2013) name: an undisaggregated SAT or ACT benchmark averages sub-scores of very different predictive validity and very different income sensitivity, so the benchmarking layer should report section-level aggregates wherever the data permit.

10 Identification of Data Gaps

Every gap below is drawn directly from the coverage numbers computed above, together with findings documented in `docs/DATA_DICTIONARY.md`, `docs/EDA_features_joined.md`, and `docs/BOB_BRIEFING.md` during the project’s development.

The first and most consequential is the **structural gap for private schools**. CRDC — the richest source, carrying AP participation, dual enrollment, and testtaker rates — is public-school-only by federal design. Section 5 shows this starkly: private schools have near-zero CRDC coverage regardless of school quality. This is not fixable by better matching, because the data simply does not exist for private schools at the federal level; only the client’s NU export fills it, and only where NU has data.

The second is a set of **vintage and freshness gaps**. Most `nu_*` fields (56 of them) carry no confirmed per-variable measurement date, only the export file’s timestamp. CRDC’s vintage is confirmed (SY2021-22) for the CRDC file itself, but that file arrives bundled inside a teammate’s gitignored export, so its freshness cannot be independently re-verified from this repo’s own history (see the 2026-07-17 update in `docs/DATA_DICTIONARY.md`).

The third is **coverage, quantified**, from Section 3: rigor-classifier signal exists for only 63.1% of the universe; complete data across all eight clustering features exists for only 16.9%; SAT coverage is 32.3%; and ACT coverage is 1.2% nationally, effectively unusable outside Illinois.

The fourth is **identifier and linkage gaps**. The `custom_id` field in the client’s org export matches no federal ID system and is not unique, making it unusable as a join key. Investigating it surfaced a separate and more serious finding: at least 62 confirmed (644 suspected) CEEB codes show a digit-shift corruption pattern, meaning some schools’ primary join key may itself be wrong.

The fifth is the **true per-pupil funding gap**. Census F-33 has no enrollment or membership field, and no verified district-level enrollment source is loaded, so the national funding feature is a population-based proxy (SAIPE school-age residents) rather than true per-pupil expenditure.

11 Assessment of Associated Risks

Table 3 records the risks these gaps create, each with its likelihood, severity, and the specific evidence behind it. None is speculative; every entry points to a quantified finding in the project’s own record.

Table 3: Risk register for the platform’s data gaps.

Risk	Likelihood	Severity	Evidence
CRDC access loss changes rigor tiers substantially	Confirmed, not hypothetical	High	52.5% of CRDC-covered schools change rigor tier if the CRDC signal is removed (<code>docs/RIGOR_CLASSIFICATION.md</code>) — quantified, not estimated.
Private schools systematically under-represented in rigor/funding features	Certain (structural)	High	Section 5; a platform meant to serve <i>all</i> applicants has a built-in blind spot for ~13,900 private high schools.
Rigor tier could reproduce socioeconomic ordering, undermining the core value proposition	Checked, found low	Low (currently)	Rigor-tier/poverty correlation is only -0.070 , versus -0.385 for raw SAT — the risk is real for outcome-style measures but empirically low for this opportunity-structure design.
CEEB corruption causes silent school misattribution	Confirmed for 62+ codes	Medium	A corrupted CEEB could attach a school’s data to the wrong record, or fail to match at all, without any error being raised.
Funding proxy misrepresents true per-pupil spending	Certain (by construction)	Medium	Population $\div$ revenue is not the same denominator as enrolled students; districts with unusual population-to-enrollment ratios are most affected.
Weak natural cluster structure over-interpreted as meaningful school “types”	Confirmed	Medium	Silhouette score 0.205 (below the conventional 0.25 threshold) — any narrative built on hard cluster boundaries would overstate what the data support.
Reproducibility gap: a new team member or grader cannot rebuild the full pipeline	Confirmed, mitigated	Low (after mitigation)	Fixed 2026-07-17 — the raw data (~2.6 GB) is documented via a shared Drive link rather than left unwritten.

12 Impact Analysis and Mitigation Strategies

CRDC dependency (High). If the Office for Civil Rights, CRDC’s steward, changes access terms — a real possibility already flagged in the project’s literature review — roughly half of currently-tiered schools would need re-scoring, and their tier could change. The mitigation already in place is that the rigor classifier logs which components were available per school (`rigor_components_available`), so a future CRDC loss is detectable and auditable rather than silent; the recommendation is to monitor CRDC’s release cadence and maintain the NU-only fallback scenario, which is already built and tested.

Private-school gap (High). Any rigor-tier or benchmarking claim about private schools rests on a thinner, less-verified foundation than the same claim about public schools — a direct threat to the platform’s stated goal of being *more* equitable than Landscape, not merely a replacement for it. The sector split is already surfaced everywhere rather than hidden inside an aggregate “all schools” number, and this report documents it explicitly; the recommendation is to state the limitation plainly in any client-facing deliverable that includes private schools, and to treat public and private as two separate populations for any statistical claim rather than one blended one.

CEEB corruption (Medium). Up to 644 schools (62 confirmed) may have data misattributed via the join key, undermining trust in any number derived from a match. The corruption is flagged via `ceeb_suspected_padding_shift` rather than silently corrected, since there is no reliable way to determine which side of a pair is right; the recommendation is to raise it with whoever owns the source export, since this is an upstream data-entry issue rather than something fixable downstream.

Funding proxy (Medium). Any per-pupil funding comparison is only as good as the population-based proxy, which could be off in either direction for specific districts. It is labeled distinctly from the true per-pupil Illinois figure via `funding_source`, so the two are never silently averaged; the recommendation is to source a true district-enrollment file (NCES CCD membership) to replace the proxy — not blocked on anyone, simply not yet done.

Weak cluster structure (Medium). There is a risk of overstating findings in the final presentation or paper if cluster assignments are treated as ground truth. The weak silhouette score is already reported as a finding about the data rather than hidden or re-framed as a modeling failure; the recommendation is that any cluster-based narrative in the final paper carry the silhouette score alongside it.

13 Insights and Recommendations

Insight 1 — opportunity-structure measures beat outcome measures on equity, and now there is data to prove it, not just cite it. SAT achievement correlates -0.385

with poverty; the rigor tier, built from AP, coursework, and test-participation *offerings* rather than scores, correlates only -0.070 . This is the central claim of the written report’s literature review (Sections 2.2 and 2.4 — Adelman, 1999, 2006; Jang and Reardon, 2019; Ho, 2020) demonstrated on the project’s own data rather than merely cited. It should lead the final paper’s Discussion, since it is a genuinely novel empirical contribution, not a restatement of prior work.

Insight 2 — the platform’s biggest limitation is not data quality, it is data existence. Most gaps identified here — private-school CRDC absence, ACT sparsity, vintage uncertainty — are not fixable by better joins or cleaner code, because the underlying federal and commercial data does not exist at the needed grain. This should be framed honestly in the final paper as a structural finding about the U.S. education-data landscape, consistent with the Section 2 literature review and with the client’s own framing (“there’s a lot of gaps in the higher-ed industry” — Bob Henkins, 2026-07-14 meeting).

Insight 3 — the platform is more reliable for public schools than private, and the paper should say so rather than presenting one blended accuracy claim. A sector-split validation section is preferable to a single aggregate metric.

Recommendation. Prioritize closing the CRDC-dependency and CEEB-corruption risks before the Week 9 presentation. Both are quantified, both have a clear owner — CRDC monitoring for whoever tracks OCR policy, CEEB corruption for whoever owns the upstream export — and both currently sit as open items rather than resolved ones.

14 Conclusion

The dataset underlying this platform is large — 53,966 schools across ten source systems — and, after the fixes made during development (LEAID derivation, the CEEB fan-out, sector misclassification), internally consistent: the Docker-backed and CSV-backed pipelines produce identical results, and 57 automated tests pass. But internal consistency is not completeness. The real constraints on this platform are structural — private schools lack CRDC by design — and are not corrigible by more engineering. The strongest finding to come out of this EDA is a positive one, and a literature-grounded one: the rigor tier is meaningfully less socioeconomically confounded than raw achievement scores, exactly the property the rigor construct was chosen for and the property the composite-indicator literature warned an offering-based measure might fail to have. The most important open risk is the CRDC-dependency finding, since half of currently-tiered schools would shift tier if that single federal source became unavailable. Both belong prominently in the final paper, not buried in an appendix.

A note on citations

The works referenced above — Adelman (1999, 2006); Geiser and Santelices (2004); Kolluri (2018); U.S. Government Accountability Office (2018); Coca et al. (2012); Bastedo et al. (2022, 2023); Jang and Reardon (2019); Ho (2020); Geiser and Studley (2002); Bettinger et al. (2013) — are the same sources reviewed in Section 2 of the project’s written report (MSDS_498_version-wk3); full bibliographic details appear in that document’s bibliography and are not duplicated here.